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Feature Encoding

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns',None)
np.random.seed(42)
print("Environment ready")
```

```
Environment ready
```

1.Nominal Encoding

In [2]:
```
df=pd.DataFrame({
    'town':['Mumbai','Mumbai','Mumbai','Mumbai','Mumbai',
           'Pune','Pune','Pune','Pune','Pune',
           'Kholapure','Kholapure','Kholapure','Kholapure','Kholapure'],
    'area':[2800,4530,6700,7640,3000,7600,2600,7000,7600,7200,3400,7800,4500,520
    'price':[4500000,6500000,1300000,2400000,670000,
            4500000,6700000,5700000,6430000,8900000,
            8700000,7800000,5600000,4200000,7800000]

})
df
```

Out[2]:

	town	area	price
0	Mumbai	2800	4500000
1	Mumbai	4530	6500000
2	Mumbai	6700	1300000
3	Mumbai	7640	2400000
4	Mumbai	3000	670000
5	Pune	7600	4500000
6	Pune	2600	6700000
7	Pune	7000	5700000
8	Pune	7600	6430000
9	Pune	7200	8900000
10	Kholapure	3400	8700000
11	Kholapure	7800	7800000
12	Kholapure	4500	5600000
13	Kholapure	5200	4200000
14	Kholapure	6100	7800000

In [3]:
```
df.head()
```

Out[3]:

	town	area	price
0	Mumbai	2800	4500000
1	Mumbai	4530	6500000
2	Mumbai	6700	1300000
3	Mumbai	7640	2400000
4	Mumbai	3000	670000

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   town    15 non-null     object
 1   area    15 non-null     int64 
 2   price   15 non-null     int64 
dtypes: int64(2), object(1)
memory usage: 492.0+ bytes
```

In [5]:
```
df_dummies_full=pd.get_dummies(df,columns=['town'],drop_first=False)
df_dummies_full
```

Out[5]:

	area	price	town_Kholapure	town_Mumbai	town_Pune
0	2800	4500000	False	True	False
1	4530	6500000	False	True	False
2	6700	1300000	False	True	False
3	7640	2400000	False	True	False
4	3000	670000	False	True	False
5	7600	4500000	False	False	True
6	2600	6700000	False	False	True
7	7000	5700000	False	False	True
8	7600	6430000	False	False	True
9	7200	8900000	False	False	True
10	3400	8700000	True	False	False
11	7800	7800000	True	False	False
12	4500	5600000	True	False	False
13	5200	4200000	True	False	False
14	6100	7800000	True	False	False

In [6]:
```
df_dummies_full=pd.get_dummies(df,columns=['town'],drop_first=True)
df_dummies_full
```

Out[6]:

	area	price	town_Mumbai	town_Pune
0	2800	4500000	True	False
1	4530	6500000	True	False
2	6700	1300000	True	False
3	7640	2400000	True	False
4	3000	670000	True	False
5	7600	4500000	False	True
6	2600	6700000	False	True
7	7000	5700000	False	True
8	7600	6430000	False	True
9	7200	8900000	False	True
10	3400	8700000	False	False
11	7800	7800000	False	False
12	4500	5600000	False	False
13	5200	4200000	False	False
14	6100	7800000	False	False

In [7]:
```
df_dummies_safe=pd.get_dummies(df,columns=['town'],drop_first=True)
df_dummies_safe
```

Out[7]:

	area	price	town_Mumbai	town_Pune
0	2800	4500000	True	False
1	4530	6500000	True	False
2	6700	1300000	True	False
3	7640	2400000	True	False
4	3000	670000	True	False
5	7600	4500000	False	True
6	2600	6700000	False	True
7	7000	5700000	False	True
8	7600	6430000	False	True
9	7200	8900000	False	True
10	3400	8700000	False	False
11	7800	7800000	False	False
12	4500	5600000	False	False
13	5200	4200000	False	False
14	6100	7800000	False	False

In [8]:

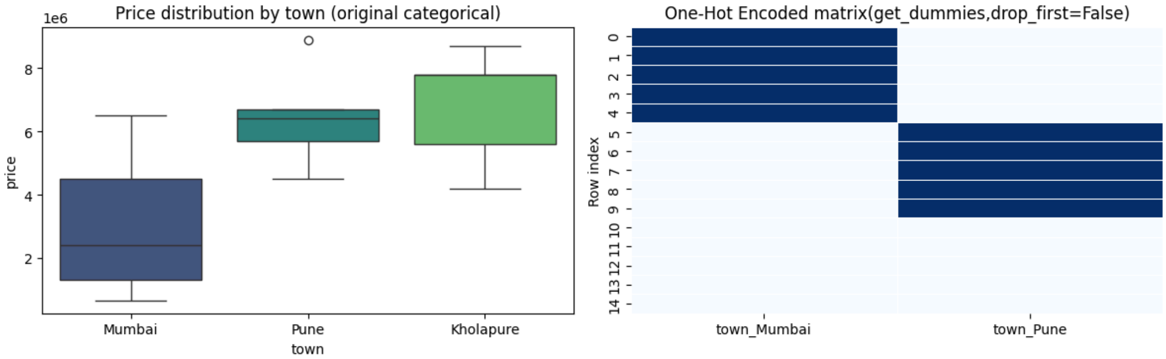
```
fig,axes=plt.subplots(1,2,figsize=(12,4))

sns.boxplot(data=df,x='town',y='price',ax=axes[0],hue='town',legend=False,palett
axes[0].set_title('Price distribution by town (original categorical)')

encoded_cols=[c for c in df_dummies_full.columns if c.startswith('town')]
heat_data=df_dummies_full[encoded_cols].astype(int)
sns.heatmap(heat_data,cmap='Blues',cbar=False,ax=axes[1],linewidth=0.5)
df_dummies_full=pd.get_dummies(df,columns=['town'],drop_first=False)
print(df_dummies_full.columns)
axes[1].set_title('One-Hot Encoded matrix(get_dummies,drop_first=False)')
axes[1].set_ylabel('Row index')

plt.tight_layout()
plt.show()
```

```
Index(['area', 'price', 'town_Kholapure', 'town_Mumbai', 'town_Pune'], dtype='obj
ect')
```



In [9]:
```
df_auto=pd.get_dummies(df,drop_first=False)
df_auto
```

Out[9]:

	area	price	town_Kholapure	town_Mumbai	town_Pune
0	2800	4500000	False	True	False
1	4530	6500000	False	True	False
2	6700	1300000	False	True	False
3	7640	2400000	False	True	False
4	3000	670000	False	True	False
5	7600	4500000	False	False	True
6	2600	6700000	False	False	True
7	7000	5700000	False	False	True
8	7600	6430000	False	False	True
9	7200	8900000	False	False	True
10	3400	8700000	True	False	False
11	7800	7800000	True	False	False
12	4500	5600000	True	False	False
13	5200	4200000	True	False	False
14	6100	7800000	True	False	False

B.One-Hot Encoding via sklearn.OneHotEncoder

In [10]:
```
from sklearn.preprocessing import OneHotEncoder

enc=OneHotEncoder(sparse_output=False,drop=None)
encoded_array=enc.fit_transform(df[['town']])
encoded_array
```

Out[10]:
```
array([[0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.]])
```

In [11]:
```
enc_df=pd.DataFrame(encoded_array,columns=enc.get_feature_names_out(['town']))
enc_df
```

Out[11]:

	town_Kholapure	town_Mumbai	town_Pune
0	0.0	1.0	0.0
1	0.0	1.0	0.0
2	0.0	1.0	0.0
3	0.0	1.0	0.0
4	0.0	1.0	0.0
5	0.0	0.0	1.0
6	0.0	0.0	1.0
7	0.0	0.0	1.0
8	0.0	0.0	1.0
9	0.0	0.0	1.0
10	1.0	0.0	0.0
11	1.0	0.0	0.0
12	1.0	0.0	0.0
13	1.0	0.0	0.0
14	1.0	0.0	0.0

2.Ordinal Encoding

In [12]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns',None)
np.random.seed(42)
print("Environment ready")
```

```
Environment ready
```

In [13]:
```
df.head()
```

Out[13]:

	town	area	price
0	Mumbai	2800	4500000
1	Mumbai	4530	6500000
2	Mumbai	6700	1300000
3	Mumbai	7640	2400000
4	Mumbai	3000	670000

In [14]:
```
df_le=df.copy()
```

In [15]:
```
df_le.head()
```

Out[15]:

	town	area	price
0	Mumbai	2800	4500000
1	Mumbai	4530	6500000
2	Mumbai	6700	1300000
3	Mumbai	7640	2400000
4	Mumbai	3000	670000

In [16]:
```
from sklearn.preprocessing import LabelEncoder

df_le=df.copy()
le=LabelEncoder()
df_le['town_label_sonu']=le.fit_transform(df_le['town'])

df_le[['town','town_label_sonu']]
```

Out[16]:

	town	town_label_sonu
0	Mumbai	1
1	Mumbai	1
2	Mumbai	1
3	Mumbai	1
4	Mumbai	1
5	Pune	2
6	Pune	2
7	Pune	2
8	Pune	2
9	Pune	2
10	Kholapure	0
11	Kholapure	0
12	Kholapure	0
13	Kholapure	0
14	Kholapure	0

In [17]:
```
df_le
```

Out[17]:

	town	area	price	town_label_sonu
0	Mumbai	2800	4500000	1
1	Mumbai	4530	6500000	1
2	Mumbai	6700	1300000	1
3	Mumbai	7640	2400000	1
4	Mumbai	3000	670000	1
5	Pune	7600	4500000	2
6	Pune	2600	6700000	2
7	Pune	7000	5700000	2
8	Pune	7600	6430000	2
9	Pune	7200	8900000	2
10	Kholapure	3400	8700000	0
11	Kholapure	7800	7800000	0
12	Kholapure	4500	5600000	0
13	Kholapure	5200	4200000	0
14	Kholapure	6100	7800000	0

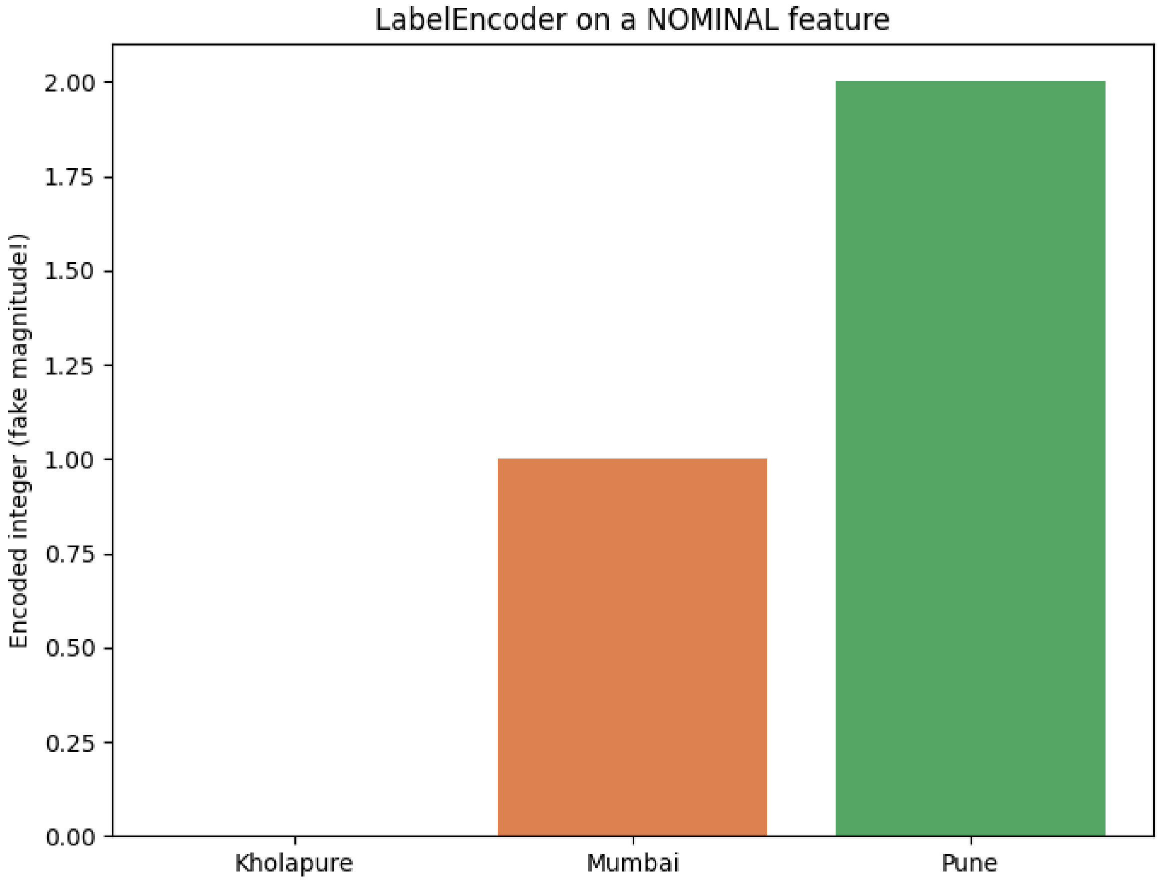
In [18]:

```
from sklearn.preprocessing import LabelEncoder

le=LabelEncoder()
df_le=df.copy()
df_le['town_label']=le.fit_transform(df_le['town'])
```

In [19]:

```
fig,ax=plt.subplots(figsize=(7,5.4))
order=df_le[['town','town_label']].drop_duplicates().sort_values('town_label')
ax.bar(order['town'],order['town_label'],color=['#4C72B0','#DD8452','#55A868'])
ax.set_ylabel('Encoded integer (fake magnitude!)')
ax.set_title('LabelEncoder on a NOMINAL feature')
plt.tight_layout()
plt.show()
```



B.OrdinalEncode

In [20]:
```
from sklearn.preprocessing import OrdinalEncoder
import pandas as pd
df_oe=df.copy()
oe_default=OrdinalEncoder()
df_oe['town_oe_default']=oe_default.fit_transform(df_oe[['town']])

edu=pd.DataFrame({'education':['Bachelors','High School','Masters','phD','Bachel
oe_ordered=OrdinalEncoder(categories=[['High School','Bachelors','Masters','phD'
edu['education_encoded']=oe_ordered.fit_transform(edu[['education']])
edu
```

Out[20]:

	education	education_encoded
0	Bachelors	1.0
1	High School	0.0
2	Masters	2.0
3	phD	3.0
4	Bachelors	1.0

3.Custom Encoding using pandas.map()

In [22]:
```
df_map=df.copy()
custom_mapping={'Mumbai':0,'Pune':1,'kholapur':2}
df_map['town_mapped']=df_map['town'].map(custom_mapping)
df_map
```

Out[22]:

	town	area	price	town_mapped
0	Mumbai	2800	4500000	0.0
1	Mumbai	4530	6500000	0.0
2	Mumbai	6700	1300000	0.0
3	Mumbai	7640	2400000	0.0
4	Mumbai	3000	670000	0.0
5	Pune	7600	4500000	1.0
6	Pune	2600	6700000	1.0
7	Pune	7000	5700000	1.0
8	Pune	7600	6430000	1.0
9	Pune	7200	8900000	1.0
10	Kholapure	3400	8700000	NaN
11	Kholapure	7800	7800000	NaN
12	Kholapure	4500	5600000	NaN
13	Kholapure	5200	4200000	NaN
14	Kholapure	6100	7800000	NaN

Handling missing

In [23]:

```
partial_mapping={'Mumbai':0,'Pune':1}
df_partial=df.copy()
df_partial['town_mapped']=df_partial['town'].map(partial_mapping)

print("Rows with unmapped category become NaN:")
df_partial[df_partial['town_mapped'].isna()][['town','town_mapped']]
```

```
Rows with unmapped category become NaN:
```

Out[23]:

	town	town_mapped
10	Kholapure	NaN
11	Kholapure	NaN
12	Kholapure	NaN
13	Kholapure	NaN
14	Kholapure	NaN

In [26]:

```
safe_mapping={'Mumbai':0,'Pune':1}
df_safe=df.copy()
df_safe['town_mapped']=df_safe['town'].map(safe_mapping).fillna(-1).astype(int)
df_safe
```

Out[26]:

	town	area	price	town_mapped
0	Mumbai	2800	4500000	0
1	Mumbai	4530	6500000	0
2	Mumbai	6700	1300000	0
3	Mumbai	7640	2400000	0
4	Mumbai	3000	670000	0
5	Pune	7600	4500000	1
6	Pune	2600	6700000	1
7	Pune	7000	5700000	1
8	Pune	7600	6430000	1
9	Pune	7200	8900000	1
10	Kholapure	3400	8700000	-1
11	Kholapure	7800	7800000	-1
12	Kholapure	4500	5600000	-1
13	Kholapure	5200	4200000	-1
14	Kholapure	6100	7800000	-1

In [ ]: